

Emotion Recognition in Music: A Multimodal Review of EEG, Acoustic Features, and Deep Learning Approaches

Stylianos Vantarakis 03121142
National Technical University of Athens (NTUA)
Athens, Greece
stelvantarak@gmail.com

Abstract—In this current paper, there is presented an integrated overview of the progress of the examination of emotional responses in music and their recognition through EEG, acoustic features and physiological data.

Index Terms—EEG, emotion recognition, music perception, audio analysis, deep learning, multimodal systems

I. INTRODUCTION

An overarching problem in the current state of art is the understanding and automatic recognition of emotional responses in music, both from performers and listeners. It is important to explore the way how musical sound interacts with human neurophysiology and how can we predict it using EEG signals, acoustic features, and physiological measurements (ECG, EMG, EDA). The goal is to build computational systems capable of predicting emotional dimensions like valence, arousal, tension, and joy. Because emotional processing in music is highly subjective and context-dependent, the current studies emphasize the necessity of incorporating both the acoustic structure of the stimulus and the neural responses it evokes.

A major unifying theme across these studies is the use of EEG oscillatory frequency bands as primary markers of emotional and cognitive processing:

- **Delta (0.5–4 Hz):** associated with internal physiological regulation, deep relaxation, and slow affective processing.
- **Theta (4–8 Hz):** linked to emotional arousal, focused attention, affective memory, and immersive music engagement.
- **Alpha (8–12 Hz):** correlated with relaxed wakefulness and internally directed focus; decreases in alpha often indicate heightened arousal.
- **Beta (13–30 Hz):** reflects cognitive effort, attention, and emotional engagement; frontal beta asymmetry is a strong indicator of valence.
- **Gamma (>30 Hz):** associated with high-level perceptual integration, emotional salience, and tension.

Band features are usually extracted using power spectral density (PSD), hemispheric asymmetry, short-time temporal windows, or time–frequency methods such as wavelets, although there are some more advanced approaches, like AT-DGNN model, that avoid completely the avoid handcrafted

descriptors and instead learn temporal, spatial, and graph-structured connectivity patterns directly from raw or minimally processed EEG. The way graph neural networks work is that they treat electrodes as nodes in a dynamically evolving brain graph, enabling biologically informed modeling of inter-regional interactions that conventional CNNs and LSTMs cannot fully capture.

To extract acoustic features, the studies use properties like attack slope, onset rate, spectral flux, rhythmic complexity, spectral distribution, and timbral richness to characterize the audio signals. More generalized audio representations like MFCCs, Mel-band energies, chroma vectors, spectral centroid, and energy envelopes are also widely used in listener-based modeling. Performer-centered studies focus on micro-timing, articulation, and expressive nuance, which more directly reflect the musician’s internal emotional state rather than the acoustic output alone.

Regarding machine learning and datasets, the research shows a clear progression from classical algorithms (SVM, k-NN, decision trees) to modern deep architectures that can model the high-dimensional, non-stationary nature of EEG. Canonical datasets such as DEAP, MAHNOB-HCI, SEED, SEED-IV, and AMIGOS appear frequently. The introduction of the MEEG dataset—which uses exclusively musical stimuli—significantly improves performance across multiple architectures. In particular, AT-DGNN, combining temporal learners, multi-head attention, and dynamic GNN layers, achieves state-of-the-art results (85–86% in valence/arousal). Performer-based studies show that even simpler models can reach exceptionally high accuracy (up to 97–98%) when combining EEG with ECG, highlighting the individualized nature of performer emotion signatures.

II. SUBJECT-DEPENDENT VS SUBJECT-INDEPENDENT RECOGNITION

A critical distinction across all is the difference between subject-dependent and subject-independent emotion recognition.

A. Subject-Dependent Models

Subject-dependent systems are trained and tested on data from the *same* individual. These models achieve:

- extremely high accuracy (often 90–98%),
- highly stable performance for each person,
- strong suitability for applications like music therapy or adaptive personal systems,
- consistent EEG patterns due to stable neural signatures within the individual.

Performer-oriented studies particularly benefit from this approach, as emotional expression during performance is idiosyncratic and internally regulated.

B. Subject-Independent Models

Subject-independent systems train on some individuals and test on unseen subjects. These models typically suffer:

- significant drops in accuracy (commonly 50–70%),
- high variability due to differences in brain physiology, musical background, emotional style, personality, and cultural context,
- difficulty generalizing to new participants without calibration.

This remains one of the most challenging open problems in EEG-based emotion recognition and motivates the use of GNNs, attention-based models, domain adaptation, and large multimodal datasets.

III. RESULTS FROM LITERATURE

	Accuracy		F1-score	
	KNN	SVM	KNN	SVM
EEG	94.09%	94.55%	0.93	0.94
ECG	13.94%	13.64%	0.12	0.11
ACO	33.86%	31.21%	0.27	0.27
EEGECG	98.11%	63.79%	0.97	0.56
EEGACO	74.24%	80.15%	0.68	0.74
ECGACO	31.79%	35.76%	0.26	0.26
EEGECGACO	73.18%	63.79%	0.66	0.56

Fig. 1. Subject Dependent Results

	Accuracy		F1-score	
	KNN	SVM	KNN	SVM
EEG	36.99%	86.86%	0.31	0.89
ECG	29.47%	14.77%	0.19	0.08
ACO	22.65%	19.74%	0.19	0.17
EEGECG	21.97%	51.07%	0.17	0.43
EEGACO	18.19%	23.37%	0.15	0.20
ECGACO	58.68%	56.79%	0.57	0.55
EEGECGACO	49.86%	51.07%	0.47	0.43

Fig. 2. Subject Independent Results

IV. MY EXPERIMENT

This section presents the complete experimental pipeline implemented in this work, including dataset preparation, feature extraction from EEG and audio modalities, multimodal fusion, classification models, and evaluation methodology.

A. Dataset and Problem Definition

The experiments were conducted using the DEAP dataset, which consists of recordings from 32 participants, each exposed to 40 audiovisual trials. EEG signals were recorded using 40 channels at a sampling rate of 128 Hz, with each trial having a duration of approximately 63 seconds. Emotional self-assessments are provided in the form of Valence and Arousal scores on a continuous scale from 1 to 9.

The objective of the experiment is formulated as binary classification problem. Valence and Arousal scores are binarized

using a threshold of 5 - later threshold 3 was used because of the imbalance of the DEAP Dataset :

- Valence $\geq 5 \rightarrow 1$, otherwise 0
- Arousal $\geq 5 \rightarrow 1$, otherwise 0

Thus, two independent classification tasks are defined: high/low Valence and high/low Arousal.

B. Data Loading and Preprocessing

All EEG data were loaded from the original .dat files provided by the DEAP dataset. EEG recordings from all participants were concatenated, resulting in a unified dataset of:

$$32 \times 40 = 1280 \text{ EEG trials}$$

with each trial represented as a multichannel time series of 40 EEG channels.

Only the Valence and Arousal labels were retained, and binary thresholding was applied as described above.

C. Train-Test Split

The dataset was split into training and testing subsets using an 80% / 20% ratio, with a fixed random seed (`random_state = 42`) to ensure reproducibility.

D. EEG Feature Extraction

1) *Continuous Wavelet Transform (CWT)*: To capture the time–frequency characteristics of EEG signals, a Continuous Wavelet Transform (CWT) was applied using the Morlet wavelet. Wavelet scales ranged from 1 to 128.

For each EEG trial and for each EEG channel, the wavelet coefficients were computed and the signal energy was calculated at each scale according to:

$$Energy(scale) = \sum_t |CWT(scale, t)|^2$$

This process results in a three-dimensional representation:

$$(N, 40, 128),$$

where N denotes the number of EEG trials. The CWT representation provides rich time–frequency information without relying on predefined EEG frequency bands and is well suited for modeling affective neural activity.

2) *Permutation Entropy*: To complement the time–frequency representation, Permutation Entropy (PE) was extracted to quantify the nonlinear complexity of EEG signals. Each EEG channel was divided into 8 non-overlapping epochs, and permutation entropy was computed for each epoch.

The resulting feature representation has the form:

$$(N, 40, 8).$$

Permutation entropy captures signal irregularity, chaos, and cognitive load, providing information that is complementary to spectral features.

E. Dimensionality Reduction with PCA

Due to the high dimensionality of EEG features, Principal Component Analysis (PCA) was employed for feature reduction. PCA was applied to:

- raw EEG representations,
- CWT-based features,
- fused EEG–audio features.

EEG features were reshaped by merging the sample and channel dimensions, standardized using z-score normalization, and projected onto the top k principal components. The reduced features were then reshaped back to:

$$(N, 40, k).$$

This process reduces redundancy, removes correlations, and improves model generalization.

F. Feature Selection Using mRMR

To further enhance discriminative power, Minimum Redundancy Maximum Relevance (mRMR) feature selection was applied. Feature selection was performed separately for:

- Valence,
- Arousal,

and independently on CWT and PCA-based EEG features.

The mRMR algorithm selects features that maximize relevance to the target labels while minimizing redundancy among selected features. The output consists of index sets corresponding to the selected feature dimensions.

G. Audio Feature Extraction

The DEAP dataset includes 36 MIDI audio stimuli, corresponding to the first 36 trials of each participant. Each MIDI file was converted to a WAV signal using FluidSynth. Audio processing and feature extraction were performed using the librosa library.

For each audio stimulus, the following features were extracted:

- Chroma features (12 dimensions), capturing harmonic and tonal information,
- Spectral flux (1 dimension), representing rhythmic and energy variations,
- Spectral roll-off (1 dimension), capturing spectral brightness.

This resulted in a 14-dimensional audio feature vector per stimulus. Audio features were aligned with EEG data by retaining the first 36 trials per participant, yielding:

$$32 \times 36 = 1152 \text{ multimodal samples.}$$

H. Multimodal EEG–Audio Fusion

Multimodal fusion was performed at the feature level. Audio features were replicated across the 40 EEG channels and concatenated with EEG features along the feature dimension. The fused representation was then projected into a lower-dimensional space using PCA with 128 components.

The final fused feature tensor has the form:

$$(1152, 40, 128).$$

This early fusion strategy integrates neural responses and acoustic properties, enabling joint modeling of brain activity and musical structure.

I. Classification Models

Both classical machine learning and deep learning models were evaluated.

1) Classical Machine Learning:

- Support Vector Machine (SVM) with RBF kernel and class-balanced weighting (in some cases),
- Random Forest with 500 trees, maximum depth of 20, and balanced class weights (in some cases).

2) Deep Learning Models:

- Recurrent Neural Network (LSTM) for modeling sequential EEG representations,
- 1D Convolutional Neural Network (CNN) for learning spatial EEG patterns.

J. Training Procedure

All deep learning models were trained using:

- Binary Cross-Entropy loss with logits,
- Adam optimizer,
- StepLR learning rate scheduler,
- 20 training epochs,
- Batch size of 64.

K. Evaluation Metrics

Model performance was evaluated separately for Valence and Arousal classification using the following metrics:

- Accuracy,
- Sensitivity (Recall),
- F1-score.

V. ANALYSIS OF EXPERIMENTAL RESULTS

The experimental results reveal several consistent trends regarding the contribution of EEG and audio modalities, the effects of class imbalance, the impact of threshold selection, and the behavior of classical and deep learning classifiers in a subject-independent setting.

TABLE I
VALENCE CLASSIFICATION RESULTS (THRESHOLD = 5, WITHOUT CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.770	0.086	0.145
RF	Audio-only	0.814	0.163	0.246
RF	Fused	0.758	0.588	0.517
SVM	EEG-only	0.773	0.000	0.000
SVM	Audio-only	0.814	0.000	0.000
SVM	Fused	0.623	0.745	0.466
RNN	EEG-only	0.758	0.000	0.000
RNN	Audio-only	0.814	0.000	0.000
RNN	Fused	0.779	0.000	0.000

TABLE II

AROUSAL CLASSIFICATION RESULTS (THRESHOLD = 5, WITHOUT CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.781	0.075	0.125
RF	Audio-only	0.753	0.510	0.477
RF	Fused	0.719	0.186	0.198
SVM	EEG-only	0.785	0.000	0.000
SVM	Audio-only	0.779	0.000	0.000
SVM	Fused	0.584	0.698	0.385
RNN	EEG-only	0.793	0.000	0.000
RNN	Audio-only	0.779	0.000	0.000
RNN	Fused	0.814	0.000	0.000

TABLE III

VALENCE CLASSIFICATION RESULTS (THRESHOLD = 5, WITH CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.762	0.259	0.330
RF	Audio-only	0.684	0.698	0.451
RF	Fused	0.749	0.588	0.508
SVM	EEG-only	0.613	0.993	0.754
SVM	Audio-only	0.390	0.884	0.350
SVM	Fused	0.623	0.745	0.466
RNN	EEG-only	0.773	0.000	0.000
RNN	Audio-only	0.814	0.000	0.000
RNN	Fused	0.779	0.000	0.000

TABLE IV

AROUSAL CLASSIFICATION RESULTS (THRESHOLD = 5, WITH CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.766	0.170	0.231
RF	Audio-only	0.697	0.745	0.521
RF	Fused	0.723	0.186	0.200
SVM	EEG-only	0.785	0.038	0.068
SVM	Audio-only	0.766	0.118	0.182
SVM	Fused	0.541	0.767	0.384
RNN	EEG-only	0.793	0.000	0.000
RNN	Audio-only	0.779	0.000	0.000
RNN	Fused	0.814	0.000	0.000

TABLE V

VALENCE CLASSIFICATION RESULTS (THRESHOLD = 3, WITHOUT CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.714	0.279	0.331
RF	Audio-only	0.779	0.605	0.515
RF	Fused	0.742	0.605	0.531
SVM	EEG-only	0.721	0.000	0.000
SVM	Audio-only	0.779	0.000	0.000
SVM	Fused	0.577	0.814	0.463
RNN	EEG-only	0.744	0.000	0.000
RNN	Audio-only	0.779	0.000	0.000
RNN	Fused	0.771	0.000	0.000

A. Overall Performance Trends

Across almost all experimental configurations, overall accuracy values remain relatively high (approximately 70–80%), while sensitivity and F1-score often collapse to near-zero values, particularly for EEG-only and deep learning models. This discrepancy indicates that accuracy alone is a misleading metric under strong class imbalance, as models frequently

TABLE VI

AROUSAL CLASSIFICATION RESULTS (THRESHOLD = 3, WITHOUT CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.725	0.186	0.234
RF	Audio-only	0.711	0.698	0.526
RF	Fused	0.703	0.256	0.267
SVM	EEG-only	0.734	0.000	0.000
SVM	Audio-only	0.742	0.000	0.000
SVM	Fused	0.558	0.744	0.394
RNN	EEG-only	0.752	0.000	0.000
RNN	Audio-only	0.742	0.000	0.000
RNN	Fused	0.768	0.000	0.000

TABLE VII

VALENCE CLASSIFICATION RESULTS (THRESHOLD = 3, WITH CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.699	0.465	0.437
RF	Audio-only	0.646	0.814	0.558
RF	Fused	0.725	0.628	0.552
SVM	EEG-only	0.577	0.977	0.719
SVM	Audio-only	0.374	0.907	0.349
SVM	Fused	0.577	0.814	0.463
RNN	EEG-only	0.744	0.000	0.000
RNN	Audio-only	0.779	0.000	0.000
RNN	Fused	0.771	0.000	0.000

TABLE VIII

AROUSAL CLASSIFICATION RESULTS (THRESHOLD = 3, WITH CLASS WEIGHTING)

Model	Input	Accuracy	Sensitivity	F1-score
RF	EEG-only	0.711	0.326	0.341
RF	Audio-only	0.665	0.814	0.561
RF	Fused	0.708	0.279	0.291
SVM	EEG-only	0.734	0.093	0.160
SVM	Audio-only	0.707	0.163	0.237
SVM	Fused	0.535	0.791	0.401
RNN	EEG-only	0.752	0.000	0.000
RNN	Audio-only	0.742	0.000	0.000
RNN	Fused	0.768	0.000	0.000

converge to predicting only the majority class. As a result, high accuracy is achieved without meaningful detection of minority-class samples.

This phenomenon is especially pronounced for SVM and RNN models, where sensitivity is frequently equal to zero despite accuracy values exceeding 75%.

1) *EEG-Only Models:* EEG-only classification achieves moderate accuracy (approximately 71–79%) but consistently low sensitivity, particularly for Valence classification. This behavior reflects the difficulty of extracting stable emotion-related neural patterns under subject-independent training. Despite the use of time/frequency features via Continuous Wavelet Transform and nonlinear descriptors such as permutation entropy, EEG-based models fail to generalize across participants. Random Forest classifiers outperform SVM and RNN models in sensitivity, suggesting that ensemble-based methods are better suited to handling heterogeneous EEG feature distributions.

2) *Audio-Only Models*: Audio-only classifiers consistently outperform EEG-only models in terms of sensitivity and F1-score, especially for Arousal classification. This is expected, as arousal is strongly correlated with acoustic properties such as energy, spectral flux, and rhythmic variation.

Random Forest audio-only models achieve F1-scores above 0.50 in multiple configurations, confirming that acoustic descriptors provide more globally consistent emotional cues compared to EEG features, which are strongly subject-dependent.

B. Effect of Multimodal EEG–Audio Fusion

Multimodal feature-level fusion generally improves sensitivity compared to EEG-only models, particularly for SVM and Random Forest classifiers. However, this improvement often comes at the cost of reduced overall accuracy.

This trade-off indicates that fused models better capture minority-class samples, leading to improved recall and F1-score, while sacrificing some majority-class predictions. SVM-based fusion models achieve the highest sensitivity values across most settings, sometimes exceeding 0.75, but often exhibit reduced accuracy due to overprediction of the minority class.

These results suggest that early fusion is beneficial but requires more advanced balancing mechanisms, such as attention-based fusion or modality weighting, to prevent dominance of one modality over the other.

C. Impact of Class Weighting

The introduction of class weighting significantly alters classifier behavior, particularly for SVM models. With class weighting enabled, sensitivity increases dramatically (reaching values up to 0.99 for Valence), while accuracy drops substantially.

Random Forest models exhibit more stable behavior under class weighting, achieving moderate improvements in sensitivity without extreme degradation in accuracy. These findings highlight the inherent trade-off between recall and precision and reinforce the importance of reporting F1-score rather than accuracy alone.

D. Effect of Threshold Selection

Lowering the binarization threshold from 5 to 3 improves class balance and results in increased sensitivity and F1-score across most models. This effect is particularly evident for audio-only and fused Random Forest classifiers, where F1-scores consistently exceed 0.50.

These results suggest that mid-range emotional states introduce ambiguity that negatively impacts binary classification when using a strict threshold of 5. Threshold selection therefore plays a critical role in perceived model performance and should be carefully justified.

E. Deep Learning Model Behavior

Contrary to expectations, RNN-based models consistently fail to detect the minority class, yielding zero sensitivity and F1-score across all configurations. This behavior can be

attributed to limited training data, lack of subject-specific modeling, and the loss of temporal structure due to aggressive dimensionality reduction.

These findings demonstrate that deep learning architectures do not inherently outperform classical methods in EEG-based emotion recognition, particularly in subject-independent scenarios with limited data.

VI. SUMMARY OF FINDINGS

The experimental results lead to the following key conclusions:

- Audio features provide more robust and generalizable emotional cues than EEG features.
- EEG-based emotion recognition remains highly individualized and difficult to generalize across subjects.
- Multimodal fusion improves sensitivity but introduces accuracy trade-offs.
- Random Forest classifiers provide the most stable performance across modalities and thresholds.
- Class weighting and threshold selection critically influence performance interpretation.
- Deep learning models fail to generalize without subject-specific adaptation or substantially larger datasets.

REFERENCES

- [1] Xiao, Minghao, et al. "MEEG and AT-DGNN: Advancing EEG Emotion Recognition with Music and Graph Learning." arXiv preprint arXiv:2407.05550 (2024).
- [2] Cui, Xu, et al. "A review: Music-emotion recognition and analysis based on EEG signals." *Frontiers in neuroinformatics* 16 (2022): 997282.
- [3] Daly, Ian, et al. "Music-induced emotions can be predicted from a combination of brain activity and acoustic features." *Brain and cognition* 101 (2015): 1-11.
- [4] Turchet, Luca, et al. "Emotion recognition of playing musicians from EEG, ECG, and acoustic signals." *IEEE Transactions on Human-Machine Systems* (2024).
- [5] Vassilis Lyberatos, et al. "Music Interpretation and Emotion Perception: A Computational and Neurophysiological Investigation" .